

HYDROCARBONS

Hydrocarbons

The compound consisting carbon and hydrogen are called hydrocarbons.

It has two types

- ✓ Open Chain Hydrocarbon
- ✓ Closed Chain hydrocarbons

Open Chain Hydrocarbons

The hydrocarbons in which 1st carbon is not linked to the last carbon are called open chain hydrocarbons

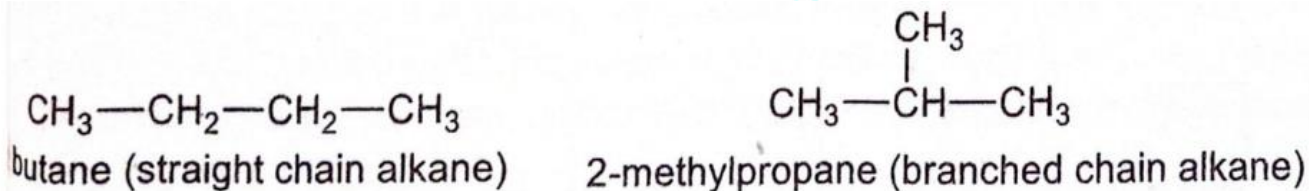
Types:

- ✓ Saturated Hydrocarbons (Alkanes)
- ✓ Unsaturated Hydrocarbons (Alkenes & Alkynes)

Saturated Hydrocarbons:

- Hydrocarbon in which all the C-C bonds are single are called saturated hydrocarbons
- All the Carbons are sp³ hybridized.
- They may be straight chain or branched chain

Examples:

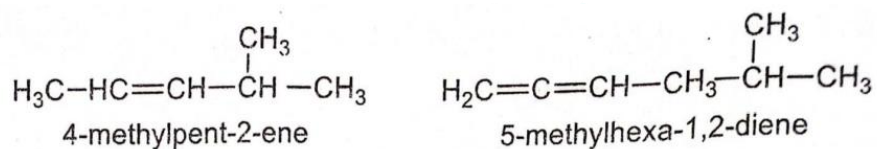


Unsaturated Hydrocarbons:

These hydrocarbons have at least one double (in alkenes) or triple (in alkynes) bond.

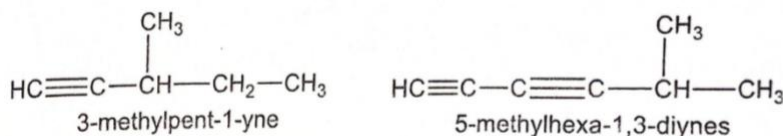
Alkenes:

These are the hydrocarbons which have at least one double bond between the two carbon atoms.



Alkynes:

These are the hydrocarbons which have at least one triple bond between the two carbon atoms.



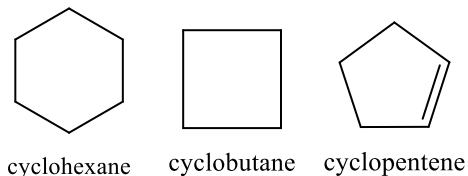
Closed Chain Hydrocarbons

Closed chain hydrocarbons have no terminal carbon in principal carbon skeleton. All carbon atoms are bonded to, at least, two other carbon atoms. They are divided into two classes.

- Alicyclic Hydrocarbons
- Aromatic Hydrocarbons

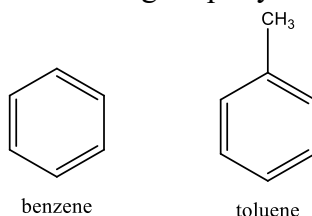
Alicyclic hydrocarbons:

The word alicyclic is blend of the two words, aliphatic and cyclic which means they are aliphatic hydrocarbons with rings of carbon atoms.



Aromatic Hydrocarbons:

Aromatic hydrocarbons have at least one benzene ring in their molecules. The simplest aromatic hydrocarbon is benzene. Each carbon of benzene ring is sp^2 hybridized.



REACTIVITY OF ALKANES

“Alkanes are also called paraffins. The word paraffins is combination of two Latin words "parum" means little and "affinis" means reactivity.

The wordly meaning of alkanes is that they are less reactive. The low reactivity of alkanes is attributed to two factors.

- ✓ Non-polar bond
- ✓ Strength of sigma bond

Non-polar bond:

- The electronegativity values of carbon and hydrogen atoms are 2.5 and 2.1 respectively.
- The difference in their electronegativity is negligible so bonding electrons are symmetrically distributed among them.
- Since alkanes are non-polar so acids, bases, oxidizing agents and reducing agents will not react with alkanes.

Strength of sigma Bond:

- Alkanes have only sigma (σ) bonds, they do not have pi (π) bonds. Sigma (σ) bond is a strong bond.
- The sigma electrons in alkanes reflects their inertness.
- Although, alkanes do not react with other reagents under normal conditions but they do react with some substances at high temperature.

Reaction of Alkanes

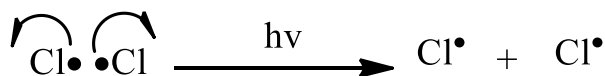
The two important reactions shown by alkanes are given.

- a. Free Radical Substitution Reaction
- b. Cracking

Free Radical Substitution Reaction

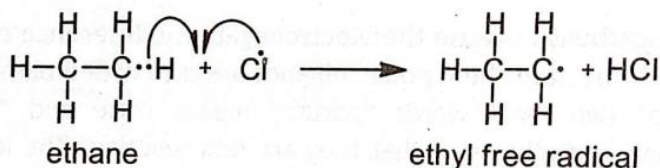
1)- Initiation:

Cl_2 split homiletically in the presence of sunlight to produce the Cl free radical.

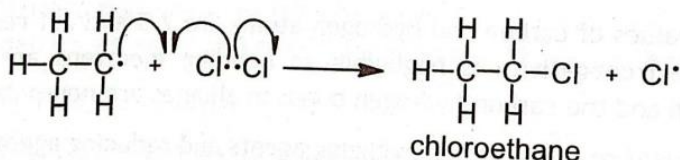


2. Propagation

As soon as the chlorine radical is formed in initiation step, it attacks on ethane molecule to give ethyl free radical.



Ethyl free radical reacts with chlorine molecule to form chloroethane and another chlorine free radical.

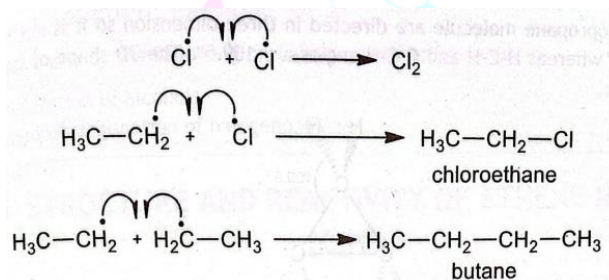


If we use excessive chlorine we get mixture of chlorosubstituted alkanes (like 1,1-dichloroethane, 1,1,1-trichloroethane, 1,1,2,2-tetrachloroethane etc).

The final product, in presence of enough chlorine, is 1,1,1,2,2,2-hexachloroethane (C_2Cl_6).

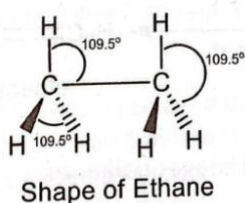
3. Termination:

In termination stage, there is possibility of reaction among different free radicals formed in propagation step.



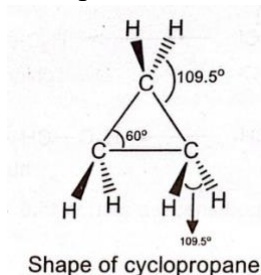
Shape of Ethane Molecules (C_2H_6)

- ✓ Ethane molecule has two carbon atoms bonded by single covalent bonds with one another. To each carbon atom, three hydrogen atoms are bonded by single covalent bond.
- ✓ When it comes to hybridization of atoms, each carbon atom in ethane molecule undergoes sp^3 hybridization.
- ✓ Each orbital shared by the two carbon atoms, either with hydrogen atoms or with one another, is sp^3 hybridized orbital.
- ✓ Ethane molecule is not planar because all covalent bonds around carbon atoms make an angle of 109.5° with one another.
- ✓ This arrangement of atoms in the space gives tetrahedral shape to the atoms around each carbon atom.



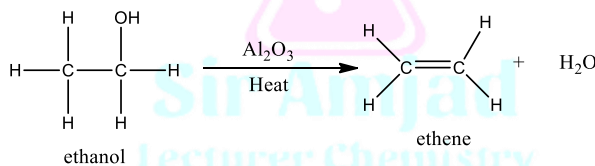
Shape of Cyclopropane Molecules (C₃H₆)

- ✓ The molecule of cyclopropane has three carbon atoms bonded by single covalent bonds with one another in the form of a ring.
- ✓ There are two hydrogen atoms bonded by single covalent bond with each carbon atom.
- ✓ In cyclopropane molecule, each carbon atom shows sp³ hybridization.
- ✓ The orbitals shared between carbon atoms or with hydrogen atoms are sp³ hybridized orbitals with tetrahedral arrangement.
- ✓ The bonds in cyclopropane molecule are directed in three-dimension so it is not planar.
- ✓ The C-C-H angles are 109.5°. The 3D shape of molecule is shown.

**Alkenes****Preparation of Alkenes****1. Dehydration of Alcohols**

Dehydration is elimination reaction which involves the removal of water. This reaction takes place by passing ethanol vapours on hot aluminium oxide powder as catalyst.

In addition, concentrated sulphuric acid (H₂SO₄) and phosphoric acid (H₃PO₄) also acts as catalysts and dehydrating agents in this reaction.

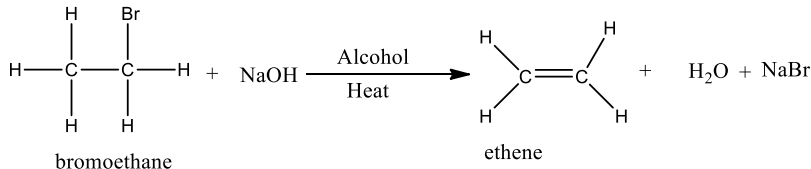
**2. Dehydrohalogenation of haloalkanes:**

Dehydrohalogenation of halogenoalkanes is the removal of hydrogen and halogen in the form of hydrogen halide molecule.

This is also elimination reaction.

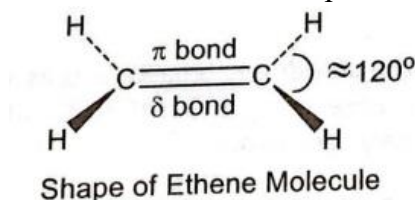
Example:

We can prepare ethene from bromoethane in presence of hot ethanolic sodium hydroxide.

**Structure and Reactivity of Ethene Molecule**

- ✓ Ethene molecule has two carbon atoms bonded by double covalent bond and two hydrogen atoms singly bonded to each carbon atom.
- ✓ A double bond consists of one sigma (σ) and one pi (π) bond.

- ✓ Each doubly bonded carbon atom shows sp^2 hybridization with trigonal planar arrangement.
- ✓ The sigma (σ) bond between the two carbon atoms is formed by head-on overlap of two sp^2 hybridized orbitals between the two carbon atoms. The two carbon atoms have strong hold on sigma (σ) electrons and this is why sigma (σ) bond is difficult to broken.
- ✓ The pi (π) bond between the two carbon atoms is formed by parallel overlap of the two unhybridized "p" orbitals on each carbon so the pi (π) electrons are weakly held by the two carbon atoms. This is why pi (π) bond is weaker than sigma (σ) bond.
- ✓ Each carbon atom in ethene molecule forms two sigma (σ) bonds with two hydrogen atoms by sp^2 -s orbitals overlap and one sigma (σ) bond with other carbon atom by sp^2 - sp^2 orbitals overlap.
- ✓ Ethene molecule is planar because all the atoms lie in the same plane with bond angles almost 120° , Such a molecule in which all the atoms lie in the same plane is called a planer molecule.



Chemical Reaction of Alkenes

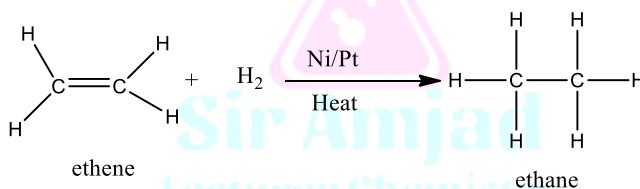
1. Hydrogenation

The addition of hydrogen gas to alkenes to form corresponding alkanes is called hydrogenation.

Conditions:

This reaction takes place at high temperature in the presence of finely divided nickel or platinum catalyst.

Examples

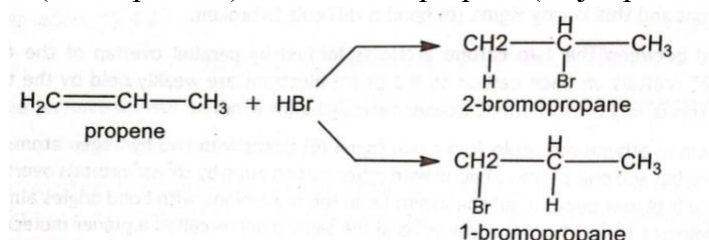


2. Hydrohalogenation of Alkenes

"The reaction in which hydrogen halide is added to alkene across double bond to form corresponding halogenoalkane is called Hydrohalogenation."

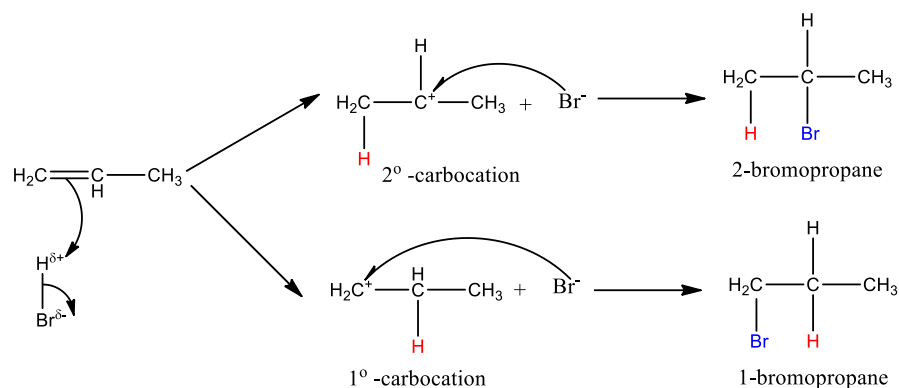
Example:

When propene ($CH_3-CH=CH_2$) is bubbled over concentrated solution of hydrogen bromide (HBr), we get mixture of 1-bromopropane (minor product) and 2. bromopropane (major product).



Mechanism:

Hydrogen atom of HBr attack on the double bond and form mixture of two carbocation, primary carbocation and tertiary carbocation.

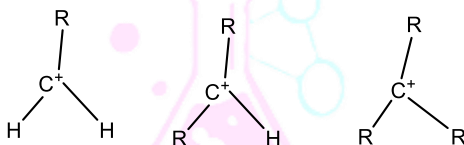
**Markonilov's Rule:**

According to this rule;

“When an unsymmetrical reagent like HBr attack on an unsymmetrical alkenes the hydrogen is attached to that carbon atom which has more hydrogen atoms”.

Stability of carbocation:

- ✓ The secondary carbocation is more stable than primary and can form more readily so the reaction favours the route of more stable carbocation.
- ✓ The stability of carbocations depends on the extent of positive inductive effect of alkyl groups. There are three types of carbocations, i. e primary, secondary and tertiary carbocations. They follow the following stability order.



- ✓ The positive charge of carbon tends to attract electrons from neighboring atoms. In addition, carbocations are also stabilized by the presence of alkyl groups.
- ✓ Alkyl groups are electron donating groups because carbon (2.5) has a higher electronegativity than hydrogen (2.1). When a carbocation is attached to one of these alkyl groups, the alkyl group will donate electrons to the positively charged carbon to stabilize the carbocation.
- ✓ This donation of electrons via sigma bonds is known as the inductive effect. As a result, the positive charge density of the carbocation decreases, making the carbocation more stable.

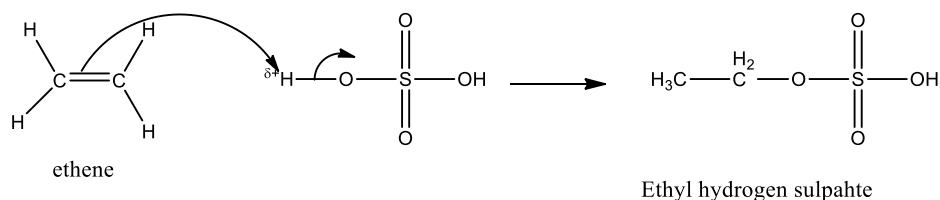
3. Hydration of alkenes:

Hydration of alkenes involves the addition of water across double bond between carbon atoms.

This is the major method of preparation of ethanol on industrial scale.

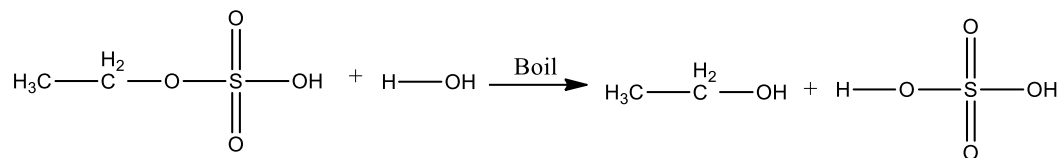
1st step:

In the first step, ethene is treated with cold concentrated sulphuric acid which gives ethyl hydrogen sulphate.



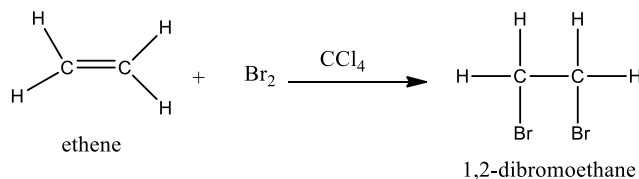
Step 2:

The mixture is heated and ethyl hydrogen sulphate decomposes to ethanol.

**4. Halogenation:**

At room temperature, ethene reacts with bromine to produce 1,2-dibromoethane.

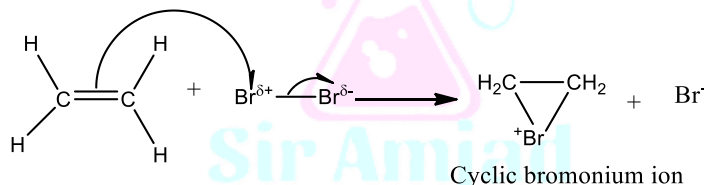
In this reaction, an inert solvent like tetrachloromethane is used.



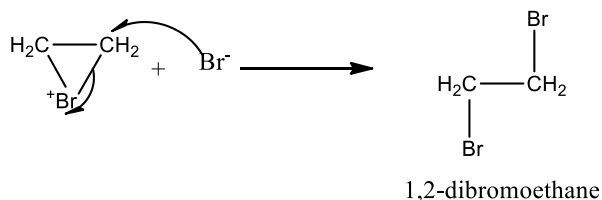
The brown colour of bromine water is discharged very fast in this reaction so bromine water can be used as a test to check unsaturation in compounds.

Mechanism:**Step 1:**

- ✓ When a non-polar bromine comes close to ethene molecule, due to repulsion between electron cloud of bromine and ethene molecule, the halogen molecule gets polarized.
- ✓ The positive part of bromine molecule acts as electrophile and attacks the pi electron cloud producing cyclic bromonium ion.

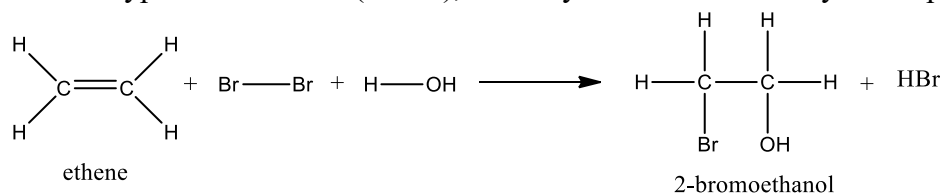
**Step 2:**

In the second step, the bromide ion (Br^-) acts as nucleophile and attacks on bromonium ion to give 1,2-dibromoethane.

**5. Halohydratation:**

Halohydratation is the addition of hypohalous acid (HOX) to alkenes across double bond.

When ethene reacts with hypobromous acid (HOBr), a halohydrin called bromohydrin is produced.

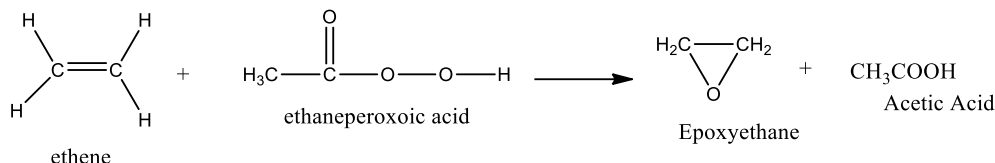


6. Epoxidation:

In this reaction, an alkene reacts with peracids, like peroxyacetic acid to form corresponding epoxide.

Example:

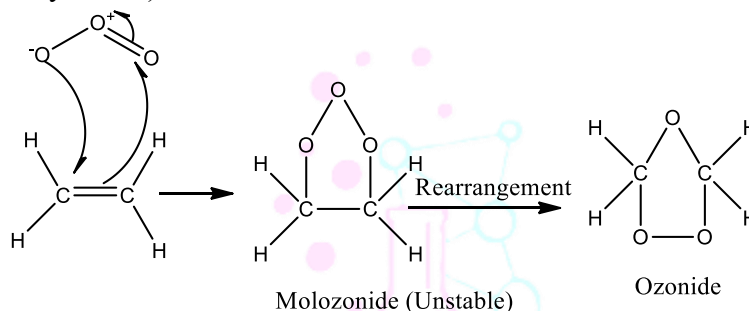
Ethene reacts with peroxyacetic acid to form epoxyethane.

**7. Ozonolysis:**

In this reaction, ozone is bubbled into very cold (-78 °C) solution of alkene in dichloromethane (CH₂Cl₂) which cleaves the double bond and form aldehydes and ketones.

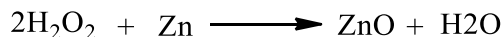
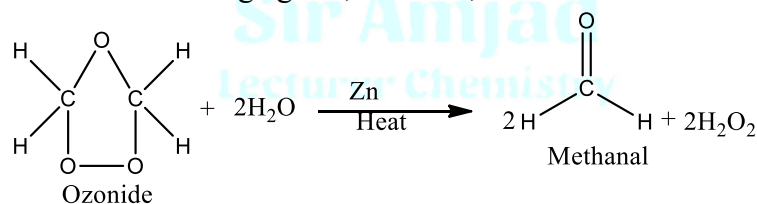
Step 1:

It involves oxygen attack on double bond to form molozonide (unstable intermediate) which rearranges rapidly to ozonide (relatively stable).

**Step 2:**

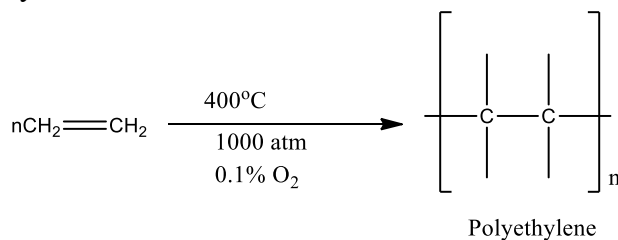
Ozonides cannot be easily isolated which means that they are not stable compounds.

When ozonide is treated with mild reducing agents, like zinc, it reduces to methanal.

**8. Polymerization:**

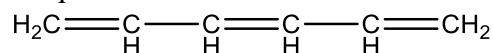
In polymerization, small molecules called monomers react together and form large molecules called polymers.

Ethene molecules can polymerize at 400 °C temperature and 1000 atm pressure in presence of traces of oxygen to form polyethene polymer.



Conjugation in Alkenes

- ✓ The word conjugation comes from Latin language which means "join together".
- ✓ Conjugation is found in those alkenes which have alternate double and single bonds. In such molecules, the carbons are sp^2 hybridized and have one unhybridized "p" orbital on each carbon on which the pi (π) electron move.
- ✓ The unhybridized "p" orbitals bridge extends till the last double bond carbon in the molecule. The double bonds in conjugated molecules are called delocalized because their pi bonds move all along on the bridge of unhybridized "p" orbitals.

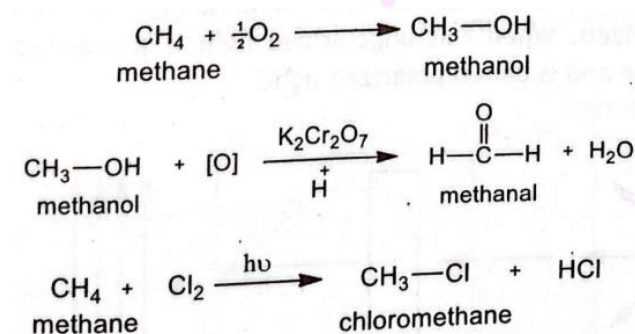


hexa-1,3,5-triene

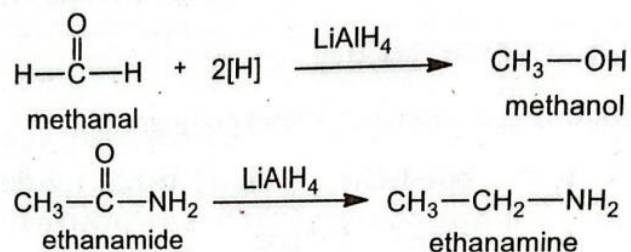
Redox reaction of organic compounds**1. Oxidation Reaction:**

- ✓ Addition of oxygen
- ✓ Removal hydrogen
- ✓ Removal of electron

Examples:

**2. Reduction Reaction:**

- ✓ Removal of oxygen
- ✓ Addition of hydrogen
- ✓ Addition of electrons



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